

SheetShift

Your rating spreadsheet, as code you own, checked cell by cell. Built with **IBM Bob**.

430,000 cells compared · 409,539 identical · 20,461 traced to 3 signed decisions · 0 unexplained

The problem

- Premiums are often computed by a **rating workbook**. An engineer re-codes it by hand for the policy system, then spot-checks a few quotes.
- The dangerous errors are rare. In our workbook a hard-coded tax and a missing credit cap each touch **1 policy in 10,000**: a 20-quote spot check misses each **99.8%** of the time.
- Python's `round()` instead of Excel's `ROUND` changes **2,552 of 10,000** quotes.
- Field audits found errors in at least 86% of the spreadsheets they examined (Panko).
- In May 2026 Washington fined an insurer \$55,000 for incorrect charges on 585 policies.

The workflow, and the Bob feature behind each step

Step	What happens	Bob feature
Map	Formulas → dependency graph, 4 units, 3 lints	Command + skill
Plan	Reads the .xlsx formulas and the PDF manual; flags disagreements	Plan mode , native document reading
Translate	4 units in parallel, every function @covers its cells	Subagents , Sheet Translator mode
Verify	430,000 cells vs the recorded workbook	PostToolUse smoke after every edit, even in subagents
Triage + briefs	Classify groups; brief each anomaly from the PDF	Sheet Triage and Sheet Analyst modes
Decide	A person chooses, citing the manual	Hook blocks Bob from the decision log
Certify	Certificate, traceability, mutation self-test	CI re-runs it on every push

Governed autonomy

- **Custom modes:** the translator edits only `service/` and `tests/`; the analyst writes only notes.
- **PreToolUse hook:** blocks edits to the workbook, grader, golden data and decision log. In a prompted guard test Bob tried a diff edit and a shell write into the grader; the hook blocked both.
- **CI is the enforcing control:** it rejects any Bob-tagged commit that touches those paths and re-hashes the grader.
- **Separation of duties:** Bob builds the service, Claude Code built the exam, a person decides.

Evidence

- Original workbook: **430,000** cells, **409,539** identical, **20,461** in rows traced to **3** decisions, **0** unexplained.
- Decision-patched workbook: **430,000 of 430,000** cells identical, so no decision hides a translation bug.
- Mutation self-test: **51 of 60** injected bugs caught; the 9 survivors are labelled by a person (8 cannot change any output, 1 input-domain gap).
- Traceability: **43 of 43** formula columns mapped to a tagged function; 5 whole-book totals declared out of scope.
- Oracle: LibreOffice 24.2 with forced recalculation, recorded; not cross-checked against Excel.
- Limits: sampled inputs, not a formal proof; no macros or volatile functions; anomalies seeded by us.

Demo

1. Bob in Plan mode reads a formula whose cached value (99) no longer matches the formula (7.5).
2. Four subagents translate the rating chain in parallel.
3. A hook's smoke check catches a translation bug mid-task; the subagent fixes it. A person decides three anomalies; Bob implements them.
4. Live: pick a workbook cell on `/trace` and see the function; run `/verify` on recorded policies.

<https://sheetshift-rho.vercel.app>

Business value

- **Buyers:** pricing and actuarial IT at carriers and MGAs, whose rates live in workbooks.
- **Why now:** AI made translation cheap. Verification is the bottleneck, and regulators ask for evidence.
- **Pricing idea:** per workbook migrated, plus per verification run in CI.
- **Different from the alternatives:** Excel-as-an-API platforms keep you on Excel; spreadsheet copilots edit the sheet. SheetShift gives owned code plus a certificate, traceability and a signed decision log.

Roadmap

- Excel cross-check at scale, next to the LibreOffice oracle
- More Excel functions and whole-book (Summary) rules
- The equivalence check as a required CI gate on every rate filing
- Batches of workbooks, one certificate each
- An offline mode that runs the translated service in the browser

Who did what

- **IBM Bob:** the rating service, its API and tests, every `@covers` tag, the translation-bug fix, the service plan, triage note and anomaly briefs, and the decisions' implementation. **4,151 of 16,861** non-generated lines. About **16** Bobcoins over **8** tasks.
- **Claude Code** (AI coding agent): the grader, the mapping tools, the synthetic workbook and manual, CI, the site shell, and drafts of the Bob pack and these slides.
- **The builder:** ran and approved every Bob task, decided every anomaly, recorded the video, submitted.

Repository: github.com/feedddosever/notmicrosoftbobdisk · Evidence: `bob_sessions/` ·

Attribution: `ATTRIBUTION.md`